5th IEEE International Conference on AI in Cybersecurity (ICAIC), 18-20 February 2026, University of Houston, Houston, United States

Jailbreaking Large Language Models: Techniques, Trends, Defenses, and Open Challenges

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2026 IEEE 5th International Conference on AI in Cybersecurity (ICAIC) | USGov | DOI: 10.1109/ICAIC67076.2026.11395786

***Abstract*—Large language models (LLMs) are rapidly becoming embedded in everyday applications, security workflows, and decision-support systems. Along with their capabilities, we have also seen a rise in techniques aimed at bypassing their guardrails. This paper presents a consolidated analysis of recent jailbreaks attacks and groups them into three high-order categories based on their underlying mechanism. Unlike prior surveys, this work introduces a mechanism-grounded taxonomy and an explicit attack-to-defense mapping that unifies cognitive, representational, and system-level jailbreak pathways. We highlight the emerging trends that make modern jailbreaks more effective and discuss where current defenses tend to fail in practice. Building on these insights, we outline practical, engineering-focused best practices to lower jailbreak risk in deployed systems. The paper closes with key open challenges—from benchmark gaps to reproducibility issues and agentic safety—that must be addressed to build more robust and dependable LLM safety architectures.**

Keywords—jailbreak, LLM, adversarial prompting, agentic systems, defense mechanisms, prompt injection, llm vulnerabilities

I. Introduction

LLMs are being deployed widely in such sensitive and high-impact domains as healthcare decision support, automated code generation, and content moderation pipelines. Alongside this progression, the security landscape of LLM-based systems is likewise changing, exposing new security issues related to misalignment of behavior in the presence of adversarial input. An increasing body of work finds that LLMs are still vulnerable to jailbreak attacks, which are well-defined prompting or interaction sequences that are designed to bypass the built-in guardrails or safety policies [1], [2]. It has been reported that even LLMs hardened with alignment strategies can be encouraged, through prompt injection, transformation-based manipulations, and context-concatenation tactics, to produce disallowed or harmful outputs [3], [4]. The closer LLMs become to tool-based agents, retrieval-augmented workflows, and long-horizon multi-turn dialogues, the wider the attack surface, and hence the possibility of more subtle, multi-step or automated "jailbreak chaining" strategies. Despite the considerable efforts made in the literature on LLM security, alignment, and privacy, most current surveys can span a wide range of issues, including model theft and data leakage, rather than narrowly focusing on jailbreak vulnerabilities and how they impact engineering [5], [6].

Consequently, practitioners and researchers are missing a compact yet structured synthesis that (a) classifies jailbreak techniques, (b) describes how their utility persists, notwithstanding alignment efforts, and (c) directs us through these findings to actual defense strategies that are relevant for real-world LLM applications [7], [8].

A. Contributions

This survey presents an organized analytical framework to comprehend jailbreak attacks that goes beyond existing compilations of prompt examples or attack anecdotes. Although prior studies have focused on discrete attack vectors, like persona steering, multilingual bypasses, or indirect prompt injection, to our knowledge, no previous survey has defined the common mechanisms that unify these attacks across linguistic, representational, and system-level boundaries. The classification of jailbreaks as Cognitive Manipulation, Representation Manipulation, and System Boundary Exploits in this taxonomy arises owing to common causal factors found in prior literature, including: (i) manipulations of the model's goal inference and cooperative tendencies, (ii) perturbations to surface forms exploiting the gap between model generalization and guardrail detection, and (iii) architectural vulnerabilities in tool-use, memory, and API interfaces. This higher-order abstraction adds conceptual clarification to existing work, and allows for comparison of why certain jailbreaks succeed across models, modalities, and deployment settings. In addition, the survey presents an Attack-to-Defense Mapping Matrix—a new decision aid that maps each attack family to the defenses with the strongest empirical mitigation potential. In addition to a summary of recent achievements in long-horizon, cross-lingual, and agentic contexts, this survey provides a system-oriented viewpoint that was previously lacking, helping to inform the development of stronger deployed LLM safety systems.

II. Background and Definitions

LLMs are typically implemented according to layered alignment and safety mechanisms, including supervised fine-tuning, reinforcement learning from human feedback (RLHF), Constitutional AI, and rule-based filtering, to limit harmful or policy-violating outputs. While these guardrails enhance the general behavioral reliability of models, alignment techniques are still susceptible to targeted manipulation, particularly if presented in adversarial prompting contexts that take advantage of model generalization tendencies or context-related ambiguity [1], [2]. In this context, jailbreak refers to any prompt, sequence, or transformation crafted to make an aligned model bypass its safety constraints and produce restricted content. Jailbreaking also differs from traditional red-teaming, which tests safety mechanisms in a risk-free manner and does not attack them as adversaries would [3]. Expanding context windows makes it possible to embed multi-staged or heavily obfuscated instructions, which compromise the effectiveness of guardrails across extended sequences [4]. External tools, retrieval pipelines, and operations-calling interfaces are all integrated or streamlined into LLMs, creating additional attack surfaces such as parameter substitution and nudged

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prompt injection [5]. Furthermore, the occurrence of multi-agent orchestration and persistent memory further exposes models to issues of long-horizon conversational drift and compounding prompt interference that gradually makes safety constraints less compliant over time [6].

III. Survey Methodology

In this work, a systematic approach is adopted to cover jailbreak research. We surveyed peer-reviewed publications, technical reports, and preprints from major security and AI venues using keywords related to ‘‘LLM jailbreak’’, ‘‘prompt injection’’, ‘‘persona attacks’’, “multilingual safety”, “tooluse exploitation”, and “LLM agents”. We focused our studies with empirical results, reproducible benchmarking, or clear attack mechanics. We then grouped attacks into recurring mechanisms using thematic coding that could then be integrated into the three-category taxonomy. lastly, Defense techniques were associated with each attack family to form the AD-Matrix.

IV. Related Studies

Recent surveys have studied jailbreaking and security in large language models, with one study offering a premature attempt at a systematic overview of jailbreaks and countermeasures, but their taxonomy is essentially exhaustive, as it fails to identify underlying causal relationships [7]. Another research extends this work to include privacy leakage, prompt injection, and misuse, providing comprehensive breadth at the cost of its relative lack of depth for linguistic, representational/abstraction, or system-induced factors contributing to jailbreak success [8]. A more recent research is focused on vulnerabilities that arise from the multimodal and agentic ecosystem and highlights the future risks with these systems and centering on modality expansion instead of the unifying and underlying failure modes across the attack families [9].

While these studies are all contributing to enhancing the ascertainment of the safety aspect of LLM, they do not present a mechanism-grounded classification of jailbreaks, or ultimately a structured mapping between attack types and the defensive layers that are most appropriate to counter them. The contribution of this survey is the introduction of several higher-order taxonomies—cognitive manipulation, representation manipulation, and system-boundary exploits—alongside an explicit attack-to-defense mapping framework that informs the more principled design of mitigation strategies.

V. Techniques: Jailbreak Taxonomy

While the jailbreak literature has dozens of individual prompting tactics, there are only a handful of such mechanisms in common. In order to establish a coherent analytic framework, this survey organizes jailbreak methods into a three-part taxonomy: Cognitive Manipulation, Representation Manipulation, and System Boundary Exploits. Cognitive Manipulation represents attacks that change the model’s goals or cooperative posture, such as persona conditioning and psychological framing. Representation Manipulation includes manipulations that change the form of the actual content (e.g., encoding, translation, invisible characters) while preserving adversarial intent in the latent space. System Boundary Exploits are the attacks intended to bypass text-level defenses by exploiting tool interfaces, function-calling layers, retrieval systems, or multi-agent workflows. The paper instantiates each category in the following subsections with the most representative methods adapted from empirical research. A mechanism-grounded taxonomy is necessary because existing jailbreak surveys largely enumerate attack variants without explaining why disparate techniques succeed across models. Grouping attacks into Cognitive Manipulation, Representation Manipulation, and System Boundary Exploits provides a principled, architecture-aware structure that captures underlying failure modes, enables clearer comparison across studies, and supports defensible mapping between attack families and appropriate mitigation layers. Figure 1 provides an overview of the taxonomy and its classifications.

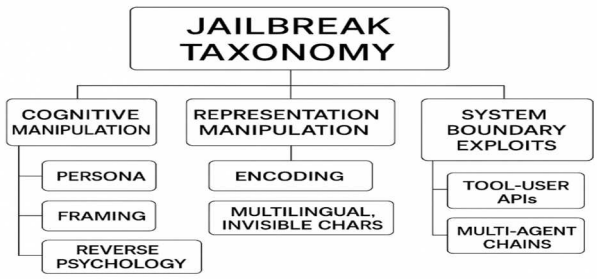

Figure 1 Jailbreak Taxonomy

A. *Persona-Based Jailbreaks*

Persona-based jailbreaks make use of an LLM’s inclination to adopt roles, characters, or identity frames that bypass or erode its existing safety constraints. Empirically, studies show that using persona conditioning changes the model’s internal preference landscape and thus boosts model compliance with malicious or prohibited requests when framed as part of a fictional or role-playing scenario. For instance, one study shows that role-based prompts have the potential to systematically undermine refusal behavior, especially when the model is invited into the role of an “unfiltered assistant,” “malicious expert,” or “obedient AI” persona, often resulting in leakage of restricted content [10], [11]. Persona overrides are also promising as long as attackers impersonate system instructions or authoritative roles, creating “identity steering” conditions that result in the model prioritizing persona consistency over safety enforcement. Other studies around cross-prompt contamination show that when a persona is present early on in a conversation, its impact can extend across multiple turns and so may lead to long-horizon jailbreaks that would not function as single-turn prompts [12].

B. *Instruction Framing and Psychological Manipulation*

Language-based instructional-framing jailbreaks use linguistic methods that take advantage of an LLM’s propensity towards cooperativeness and goal-fulfilling impulse, instead of low-level perturbations. Empirical investigations into jailbreak prompt corpora and attacker behavior show that such prompts framed as research, humor, or role-play systematically weaken safety filters compared with direct requests [13], [14]. Reverse-psychology patterns and moral or emotional appeals, like saying that harmful

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instructions are required to "prevent" an attack or to "audit security," are often seen by red-teamers as effective tactics to extract blocked material [14]. Yu et al. (2024) systematize these prompts into identifiable patterns, such as disguised intent, defined personas, imagined scenarios, and "opposite mode" instructions that ask the model to act against its usual constraints. Recent work explicitly links these patterns to mechanisms from cognitive psychology: foot-in-the-door style multi-step requests and self-persuasion framing significantly boost attack success rates through slowly changing the model's internal "goal" from safety compliance to task completion [15], [16]. Closely related are "safe transformation" framings, where the user asks the model to explain, translate, or critique harmful material rather than generating it from scratch. Analysis of jailbreak benchmarks and attacker-generated prompts has shown that the reframing often evades keyword-based or intent-based filters since the task appears analytical, even though the underlying content exposes the model's latent harmful knowledge [13], [16]. Together, these methods demonstrate weaknesses in alignment that are primarily based on surface-level intent detection and do not rely on deeper semantic and discourse-level reasoning about user goals.

C. Transformation-Based Jailbreaks

Transformation-based jailbreaks alter the representation of harmful content by translation, encoding, paraphrasing, and formatting, so that safety components cannot detect it while the base model still reconstructs the underlying intent. Yu et al. (2024) demonstrate that "structured response" patterns, including enforced language translation and program-format outputs, constitute one of the most prevalent and effective jailbreak strategies in the wild. The ciphers and non-natural encodings are an extreme case: two studies show that models learn quickly to use ciphered representations (e.g., Walnut53, Endspeak) and can be induced to decode and answer harmful questions even when the surface text appears benign, achieving near-perfect attack success rates on standard harm benchmarks [17], [18]. GitHub. +1. Multilingual transformations constitute another robust vector: Another study presents similar findings in multi-commercial and open-source models: jailbreak success is substantially higher in low-resource languages than in English [19], suggesting that alignment and filtering are predominantly English-centric. Recent work pushes this notion to the limit of imperceptible transformations.

Gao et al. develop "imperceptible jailbreaks," whereby invisible Unicode variation selectors are inserted into malicious prompts; the text appears unchanged to a user, while the text tokenises differently, giving rise to high success in jailbreak and prompt-injection attacks against multiple aligned models [20]. Other representational attacks leverage the model's differential treatment of formal formats, such as code, graphs, JSON, or XML. Adding harmful instructions in structured representations, and asking the model to "analyse" or "execute" them, has been found to have much better performance than plain-text prompts, because safety filters consider the content as inert data rather than as a user request [13], [21]. Summarization-based and compression-based encodings follow the same principle: harmful material is compressed, obfuscated, or metaphorically represented in earlier turns and later expanded on demand [16], [21]. Taken together, this class of attacks is based on the fundamental discrepancy between model-level generalization, which remains robust to many transformations, and guardrail components acting on surface forms or single-turn intent classification.

D. Multi-Turn Coercion and Conversational Drift

Multi-turn jailbreaks take advantage of the temporal dynamics of dialogue, where each response conditions the next, and safety instructions gradually lose their effect relative to local conversational goals. One study demonstrates that multi-turn "context fusion" attacks can far surpass single-shot prompts: by interleaving benign and harmful prompts, adversaries can build up context clues that, together, tend to pressure the model into unsafe completions while avoiding overt violations of policy at any single step [22]. Other studies from the literature show further that carefully designed multi-turn strategies, like asking for generic scaffolding first, then progressively specializing towards a more damaging target, result in considerable reductions in the refusal rates for GPT-4-class models, particularly when the model is encouraged to maintain consistency with its earlier, seemingly benign outputs [23], [24]. Behavior like this is deeply entrenched in the phenomenon of contextual drift and erosion of constraints. After giving some technical knowledge or acknowledging a hypothetical scenario, refusals also become far less likely when the user treats follow-up requests as clarifications or continuations of what the model already said [14], [25]. Explicit conversation memory or summarization-driven systems add another attack surface: adversaries can sow harmful or misleading seeds at the beginning of such a conversation, then turn to summarization components to safeguard them as high-level "facts" and then refer back to these internalized elements in an attempt to explain unsafe outputs [26], because most safety classifiers work at the utterance level rather than over entire histories, these slow-burn attacks often evade guardrails even when individual turns are innocuous. Multi-turn coercion thus constitutes an especially challenging class of jailbreak as the attack mechanism is distributed over time, associated with legitimate interaction, and exacerbated by the model's need for coherence and helpfulness.

E. Tool-Use, API, and Function-Calling Exploits

As LLMs are turned into tool orchestrators like code interpreters, retrieval APIs, databases, robotics interfaces, or plug-ins, the jailbreak surface goes far beyond text generation and into software behavior. One study first formalized indirect prompt injection for real LLM-mediated applications by showing that attacker-controlled content embedded in web pages or documents can override system prompts, exfiltrate data, and trigger unwanted API calls when queried by an LLM agent [27]. Later work generalizes that to multi-step agent workflows, with some demonstrating how topic-transition and reply-chaining attacks can embed adversarial instructions through intermediate messages, inducing wrong notifications for downstream tools, without the user's need [26]. In function-calling and tool-augmented systems, jailbreaks are often targeted at the tool invocation prompt or schema, rather than the natural-language interface of the model itself. One associated work on tool-invocation prompt

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hijacking found: (i) that specially crafted natural language inputs can guide the model to emit API calls with malicious arguments in the environment, (ii) take advantage of very strict JSON/XML schemas, or (iii) return malformed outputs in ways that set the downstream parsers up for a denial-of-service [20], [28]. Code-generating scenarios further increase vulnerability: adversarial requests can lead seemingly "safe" code-assistant settings to spawn exploit proofs-of-concept or insecure code configurations, even when harmful requests are blocked [29]. Multi-agent architectures compound these issues, with an agent's output becoming another agent's input; recent security studies have demonstrated that psychological and social-influence strategies can spread throughout agent collectives, with emergent jailbreak dynamics exhibiting the absence of clear-cut maliciousness [7], [16]. Tool-use jailbreaks thus bridge classical prompt manipulation and software exploitation as both point to LLM safety and system security being a coupled problem rather than single layers.

F. Trend Summary

One of the most fully quantified cases of crosslinguistic deficit is revealed. The work "Low-Resource Languages Jailbreak GPT-4" shows that translating unsafe prompts composed in English into low-resource languages results in harmful outputs: GPT-4 -4 disallowed response rates in 79% of the evaluated cases through AdvBenchmark, compared to less than 1% when the prompts were submitted using English [30]. This gap reveals a substantial structural deficiency: safety alignment is still deeply entrenched in English, and multilingual or code-switched prompts can sidestep guardrails. There is a second trend supported by quantitative evidence on the performance of multi-turn jailbreak attacks. Another paper reported a systematic study on more sophisticated multi-step dialogue attacks to popular models, including GPT-4, Claude and Gemini, that demonstrated automated multi-turn attack agents can achieve success rates of over 70% under settings where single-turn robustness is considered [31]. This indicates that safety mechanisms, which examine each prompt independently, may fail to identify adversarial strategies spread over multiple conversational steps.

These quantitative results align with the qualitative trends reported by prior surveys and red teams in general. Models with larger context windows, as well as multilingual coverage and agent-like capabilities, are more susceptible to long-horizon attacks, cross-lingual prompting, and indirect instruction chains. Thus, as LLMs become increasingly integrated with retrieval systems, memory modules, and external tools, these vulnerabilities become more significant, suggesting that jailbreak success results from the structural evolution of contemporary deployment paradigms within LLMs rather than an independent, prompt-level vulnerability. Table 1 provides a summary of the different attack techniques.

Table 1 Techniques Summary

Technique	Example	Mechanism
Persona-Based	"Act as an unfiltered AI/expert…"	Role-conditioning overrides safety by prioritizing persona consistency.
Instruction Framing	Framing harmful tasks as research, fiction, or translations	Reframing weakens intent detection; it exploits cooperative bias.
Transformation-Based	Encoded prompts (Walnut53, Endspeak), JSON/XML wrappers, low-resource language translation	Alters surface form to evade filters while preserving recoverable meaning.
Multi-Turn Coercion	Gradual scaffolding: safe steps → harmful target	Distributed intent avoids single-turn triggers; exploits contextual drift.
Tool/API Exploits	Indirect prompt injection via webpages; malicious API arguments	Attacks tool interfaces rather than text; bypasses LLM guardrails.
Emerging Trends	Cross-lingual prompts; multi-agent jailbreaks	Structural gaps in multilingual alignment; compounded vulnerabilities in agent workflows.

VI. Root Causes Behind Jailbreak Vulnerability

Many systemic and architectural obstacles explain factors that led to the unrelenting jailbreak attacks on LLMs, even when the most powerful alignment algorithms are put to use. An important reason is due to the mismatch between the model pretraining objective and the safety-layer constraints that the model is learning from, and most of the time, foundation models are trained using next-token prediction on large corpora with malicious, biased or unsafe patterns [1]. Alignment methods like RLHF or Constitutional AI place behavioral constraints only after pretraining, meaning safety layers are considered as a relatively thin post-hoc adjustment to baseline behaviors. As characterized in foundational alignment literature, this creates a disconnect between the model's learned general-purpose capability and external guardrails, allowing adversarial prompts to re-ignite latent unsafe behaviors when presented [1]. The second root cause is the model's tendency to overgeneralize and its pro-cooperative reasoning, which leaves it vulnerable to adversarial reframing. Models trained on jailbreak corpora and attack taxonomies tend to default to the expectation of user intention as cooperative, thus, as much as possible, seeking to meet requests even if they contradict safety instructions at the system level [13], [14]. Such a cooperative bias results from pretraining (liking linguistic cues) and RLHF (which rewards helpfulness), both of which cause cleverly-structured prompts, fictional scenarios, or safe-transformation tasks to bypass explicit intent filters [32].

A third issue is the sensitivity of aligned models to contextual drift over long prompts and multi-turn interactions. We build history by talking things through, and models favor consistency with prior turns. Recent empirical studies of

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multi-turn jailbreaks find that safety behavior deteriorates after a period of extended interaction, whereby the automated multi-step agents reach >70% success rates, because the initial guardrails are gradually degraded [31]. This is analogous to the integration of summarization-based memory, which remains to be integrated into many LLM systems. When harmful, misleading context is presented early, memory modules can process such cues; as such, a later request is misinterpreted by safety assessments as a benign continuation [26]. Lastly, flaws in guardrail classifiers give a large room for error for evasion. Input filters tend to rely on surface-level pattern recognition or English-centric intent detection. Cross-lingual jailbreak evaluations have shown that GPT-4 and similar models jump from <1% bad outputs in English to 79% in low-resource languages when tested against the same bad prompts [30]. Safety classifiers also find it challenging to learn harmful intent when it is embedded among encodings, transformations, or formal structures, such as JSON or code. These constraints demonstrate that guardrails are not capable of taking full advantage of the generative flexibility of modern LLMs and lack semantic depth and multilingual robustness.

VII. Best Practices to Reduce Jailbreak Success

Preventing jailbreak attacks demands a multi-layered and adaptive approach, comprising input hygiene, conversational controls, layered guardrails, tool/agent sanctions, dynamic monitoring, and regular adversarial testing. Because jailbreaks leverage systemic properties of LLMs, defense should be achieved through a distributed system architecture and not at a single layer. Comprehensive surveys of jailbreak attacks and mitigation strategies highlight the necessity of such a holistic approach [7].

A. Input and Prompt-Level Hygiene

One of the most important of these fundamental practices is to restrict the freedom of the user's inputs by using structured or schema-constrained prompts instead of unrestricted natural language. Constraining the prompts to fixed templates or controlled schema considerably restricts the attacker's ability to embed hidden or reframed harmful content. Some advice from the security community about prompt-injection states that you should sanitize the user input: remove any metadata (HTML tags, hidden markup, etc.), either reject or normalize suspicious encodings, and disallow unprocessed dumping of user-provided text without inspection. [33]. Complementary to input hygiene is "meta-prompting"; a system-prompt with high-safety awareness embeds explicit instructions on refusal criteria, critical self-review, or safety-compliance prior to any user content processing. Previous studies of aligned LLMs demonstrate that interposing such internal self-critique or policy-reminder instructions in layers results in lower classes of jailbreak success, especially those relying on naive instruction reframing [7], [34], but those defenses are less effective when attackers employ more nuanced tricks, including encoding harmful content, working with non-English or low-resource languages, or passing requests off as transformation tasks (translation, summarization, evaluation). Empirical measurements show that many jailbreaks survive such sanitization and meta-prompt defenses [33].

B. Interaction Hygiene and Multi-Turn Controls

Multi-turn interaction is one of the most reliable predictors of jailbreak success. In contrast to single-turn prompts that are widely handled by aligned models, prolonged dialogues provide the potential for a de-escalation of safety constraints. One case study illustrates this fact that automatic multi-turn attack agents performed with >70% success rates on state-of-the-art models (e.g., GPT-4, Claude, Gemini) under the Strong REJECT benchmark [31]. These results emphasize that multi-turn safety is much weaker than single-turn performance, even in very sophisticated systems. To combat this tendency, systems should enforce "safety checkpoints" in conversations. After a given number of turns, or whenever user intent appears ambiguous, the system should (1) restate safety policies, (2) reinstate the system-prompt constraints that have drifted away, and (3) trim, compress, or reset context queues to avoid harmful accumulation [31]. When employing LLMs with persistent memory or summarization modules, memory sanitization is mandatory; potentially harmful or ambiguous content should not be written in long-term memory summaries. Without these mechanisms, the systems find themselves opening up to contextual drift, where the model tends to follow conversational clues rather than general safety guidelines more and more. According to empirical studies on contextual poisoning, harmful content integrated into the dialogue's narrative structure decreases refusal rates significantly because the model emphasizes coherence more than adherence to constraints [6].

C. Layered Guardrail Architecture

Single-layer safety mechanisms—such as keyword filters or separate toxicity classifiers—only provide partial protection. Adversarial analysis consistently demonstrates that current jailbreak approaches use linguistic transformations, encoded representations, multilingual attacks, and multi-turn escalation to avoid being blocked by guardrails. For this, many current studies show the requirement for the use of many guardrails that, in turn, are used as defense-in-depth [7], [35]. A common layered architecture may include:

- Input sanitization and schema enforcement to limit how user input is organized.
- Surface filters to catch clear harmful intent.
- Semantic or classifier-based detectors with the ability to interpret transformed or paraphrased content.
- LLM-internal policy-critique modules, such as constitutional or self-audit methods.
- Last output-level safety scorers—the last obstacle to returning content back to the user.

There is some empirical evidence that such layered designs work. The JBSHIELD-based defense of combining adjustments in hidden representation with output filtering has successfully decreased average attack success for various jailbreak categories from 61% to approximately 2% in controlled experiments [36]. These results are primarily valid for open-source LLMs but show that bigger architectural changes are more effective than shallow filters. An encouraging direction in this area is unlearning-based defense, such as the SafeLLM framework, which gradually removes harmful behaviors and hidden pathways from the

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model's weights. Initial results indicate large drops in jailbreak performance for a number of open-source models without extreme performance drops [37]. Unlearning, while experimental, works on the root causes of unsafe behavior rather than treating symptoms, suggesting a more durable approach to safety. Still, multi-layered defenses are not infallible. Studies of adaptive attack have shown that attackers can combine a variety of techniques, i.e., multilingual encoding, multi-turn escalation, and tool invocation, to break through even multi-level systems [33], [38]. So layered defenses need constant retraining and updating.

D. Tool and Agent Safety Controls

As LLMs perform more and more as agents, running code or fetching data, making API calls or controlling external systems, the attack surface widens far beyond just text generation. Several empirical audits reveal that indirect prompt injection—the act of executing an attacker's controlled text via external tools that manipulate the model-is one of the primary failure modes for agentic systems [39]. These risks are mitigated in many ways by strict argument validation, where no unsafe or malformed inputs could potentially be used by downstream tools; tool whitelisting, where execution can be limited to a narrow set of safe capabilities; sandboxing, where tool execution can be isolated; and hard separation between user content and system commands. Agent "safety checkpoints" are additionally recommended in multi-agent scenarios where the safety module checks if an action aligns with policy before any such agent takes an action. One research shows that role-isolated memory and per-agent constraints effectively alleviate the cross-agent propagation of harmful instructions in multi-agent LLM ecosystems [40].

E. Monitoring, Fail-Safes, and Adaptive Defense

Not even the strongest jailbreak protections are completely immune to jailbreak. Thus, dynamic check and fail-safe mechanisms are crucial in a robust scheme. Runtime attack detection, such as abnormal API requests, strange language switching runs, encoded payloads, or multi-turn escalation, can detect attacks escaping from static technology. Audit trails can be used for post-incident examination and compliance oversight. A proactive goose-defense method such as ProAct has recently been implemented to produce "spurious safe-looking responses" in an effort to deceive automatic jailbreak-search agents, which have shown to reduce overall attack success by up to 92% under controlled adversarial simulations [41]. The use of such deception must, of course, be handled cautiously, but the deception serves as a reminder of adversarial unpredictability as a form of defensive architecture.

F. Operational and Governance Measures

But technical defenses are not sufficient in terms of maintaining safety over the long run. Routine red-teaming cycles, including adaptive attack agents, multilingual evaluation, tool interaction tests, and multi-turn simulations, are required to identify vulnerabilities as they arise. Cross-lingual testing is crucial, given that models such as GPT-4 have proven harmful output rates climbing from under 1% in English to 79% in low-resource languages when fed identical harmful prompts [30]. Strong governance frameworks, monitoring and surveillance of usage, human oversight for high-risk queries, and robust escalation procedures also mitigate operational threats. Table 2 provides an overview of the different attack, defense, and mitigation approaches.

Table 2 Attack Defense Matrix

ATTACK CATEGORY	DEFENSE TYPE	MITIGATION STRATEGY (Specific Measures)
Cognitive Manipulation (Persona, Framing, Reverse Psychology)	- Intent classification - Persona violation detection - Refusal scoring models - Self-critique LLMs	- Enforce *persona-reset* every turn; strict persona boundaries - Disallow role assumptions that contradict safety system messages - Add *counter-framing detectors* for disguised harmful intent - Foot-in-the-door pattern detection for multi-step persuasion
Representation Manipulation (Encoding, Multilingual, Invisible Unicode, Structural Formats)	- Input sanitization - Unicode normalization - Translation normalization – Structural content validators	- Strip or normalize Unicode variation selectors - Force translation into English before safety scoring (fixes multilingual bypass) - Reject unknown encoding patterns (Walnut53, Endspeak) - Require schema-validated JSON/XML to prevent hidden hostile payloads
Transformation -Based Attacks (Ciphering, Paraphrasing, Structural Wrappers)	- Semantic filters - Embedding-based safety classifiers - Retrieval-augmented refusal models	- Use *semantic similarity checks* to detect encoded harmful content - Enforce policy checks *after* decoding or translation - Apply guardrails to "explain/translate/critique" tasks, not only generation
Multi-Turn Coercion & Conversational Drift	- State-level safety scoring - Long-horizon consistency checks - Memory sanitization	- Apply "safety checkpoints" every N turns - Prune or reset history when risk increases - Prevent persistence of early-seeded harmful content in memory modules - Detect contextual drift patterns and apply forced refusal
Tool-Use & API Exploits (Indirect Prompt Injection, Malicious Arguments)	- Tool-call validation gates - Schema-hardened function calling - Sandboxed execution	- Strict argument validation for all tool calls - Separate user content from system-level code - No free-form natural language passed into APIs - Multi-agent safety checkpoint before any agent forwards outputs

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Multi-Agent Jailbreaks (Role propagation, Influence diffusion)	- Per-agent policy enforcers - Role-isolated memory - Inter-agent safety filters	- Prevent agents from interpreting other agents' outputs as direct instructions - Enforce delegation rules with constraint layers - Apply a "critical agent" to validate high-risk agent chains
Cross-Lingual Jailbreaks (Low-resource languages)	- Language detection - Forced English translation - Multilingual alignment training	- Automatically translate any non-English request before safety scoring - Expand safety alignment datasets for low-resource languages - Reject unsupported languages rather than producing unsafe output

VIII. Current Defenses and Their Limitations

The current forms of jailbreak defenses against attacks can broadly be divided into five groups: prompt hardening, input and output filtering, constitutional or self-critique techniques, safety-scoring models, and red-teaming systems. While each of these is useful for enhanced robustness, it is shown that none of these defenses work alone, and specifically, they are not sufficient with regard to adaptive or multi-turn attacks, as empirical studies all show. The prompt hardening techniques rely on a combination of resistance to system instructions by embedding refusal rules, meta-guidance, or safety reminders directly into the system prompt. Although these interventions provide modest improvements, they are often defeated with reframing, multilingual prompts, transformation-based attacks, or contextual drift through more extended conversations. Analyzing adversarial attack corpora indicates that even large-scale system-prompt engineering is susceptible to indirect prompt injection and multi-step reasoning traps [27]. Output filtering, via keyword checks, toxicity classifiers, semantic intent models, etc., offers another line of defense. But the layers cannot handle encoded, rephrased, or low-resource language prompts. GPT-4's cross-lingual jailbreak study shows this even more emphatically: while the likelihood of producing harmful outputs went from < 1% in English to 79% in low-resource languages when surfaces changed and underlying semantic content was unchanged [30]. Classifier-based filters also show weakness when attackers embed harmful intent in fictional, hypothetical, or chain-of-thought structures.

Constitutional AI and LLM-based self-critique mechanisms provide a greater sense of flexibility in safety reasoning, enabling a model to critique and edit unsafe responses in a less rigid fashion. These are still capable of degrading against (single-turn) direct requests, but are still prone to degradation for multi-turns, and are vulnerable to advanced jailbreaks such as translation attacks, cipher encodings, and multi-turn scaffolding [1]. The well-known safety-scoring models offer a last gate before content is delivered, but also can be countered when adversarial prompts take advantage of tool-use pathways, indirect context injection, or encodings. Finally, red-teaming frameworks illuminate vulnerabilities but can't provide full coverage; attackers often discover novel jailbreak patterns quicker than defensive data points can adapt to. This set of parameters highlights that multi-layered, adaptable, and continuously reviewed defense architectures are essential.

IX. Open Challenges and Future Directions

The trustworthiness of jailbreaking defenses is still being hampered by a number of unaddressed challenges, despite the fact that LLMs are currently evolving towards more sophisticated, tool-oriented, and multilingual use cases. One underlying cause is the absence of shared valuation metrics. Existing adversarial suites are highly diverse in terms of linguistic coverage and task architecture, which results in fragmented benchmarks from where only partial resilience of the model can be inferred. Recent multilingual stress tests demonstrate that vulnerabilities do not emerge evenly across languages, which only serves to underscore the need for global rather than English-centric scores [30]. Another limitation is the robustness of attack and defense results. Because of the impact of small deviations in conversation history, decoding parameters, or even environment on model behavior, jailbreak success is notoriously variable and consequently complex both in terms of scientific reproducibility and the guarantee of safety. This sensitivity suggests that latent failure modes can arise not only from processing of the prompt itself, but also from how model reasoning varies. A problem that is becoming important consists of the study of LLM-agent ecosystems. When models schedule responses, or interpret the agents' processed information, or work together collaboratively across agents, vulnerabilities can percolate through system components in an often hard-to-trace and contain manner. Empirical evidence indicates the existence of incorrect behaviors when subtle changes on a tool artifact or intermediate message can take place, despite textual guardrails [39]. The final one is a strategic challenge to protect yourself in the long term. Attack vectors evolve rapidly, and the cost of replicating adversarial scenarios is high since reviews also need to include different languages, modalities, or system settings. Having standardized multilingual benchmarks along with reproducible evaluation pipelines and safety frameworks for tools, memory, as well as agentic behavior would need to be a priority when it comes to future research.

X. Conclusion

Jailbreak attacks are a long-term and evolving threat to the secure deployment of LLMs. As this survey has illustrated, these weaknesses are not simply a result of one particular vulnerability but rather from inherent traits of current LLMs: huge context windows, flexible reasoning capabilities, and multilingual generalization increasingly connected to commercial resources and external systems. Modern defenses, including fine-grained hardening to multi-stage guardrails, afford non-trivial mitigations but are insufficient against adaptive attackers who engage across multiple turns and languages. They need common evaluation standards, reproducible testing pipelines, and safety frameworks that go beyond text generation to deal with tool use, memory, and agent coordination. Further progress will instead require continued research, adversarial attention, and system-scale design principles that can help create resilient, scalable, and trustworthy LLM deployment.

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